



# Data Converters

## Chapter 05 – Sampling Circuits

Summer Term 2026  
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## Acknowledgement

Concepts & slides are partially based on the lecture

### VLSI Data Conversion Circuits

held by

**Prof. B. Murmann, PhD, University of Hawai'i**  
**during his sabbatical at the University of Freiburg in 2009**



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## Overview

- **Introduction**
- Nonidealities
- Bottom Plate Sampling
- Design Considerations

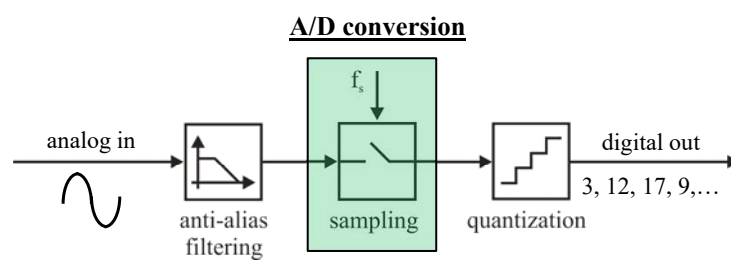
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## Introduction



How to implement a „sampling“ circuit?

Ideal Dirac sampling is infeasible!

→ There is no switch that samples a signal in „zero“ time.

**Solution: Track & Hold**

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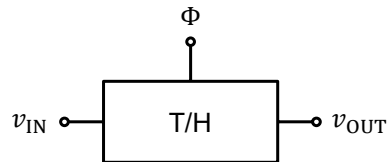
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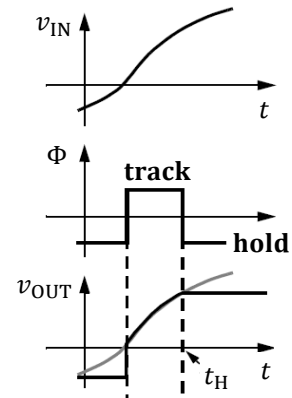
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## Ideal Track & Hold



Although inaccurate, the circuit is typically called *sample & hold* instead of *track & hold*.



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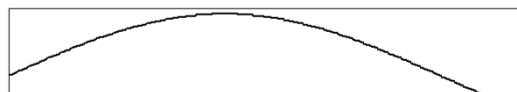
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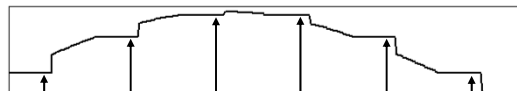
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## Nomenclature

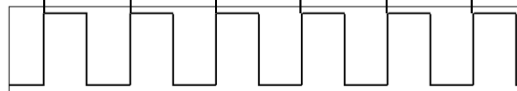
continuous-time signal



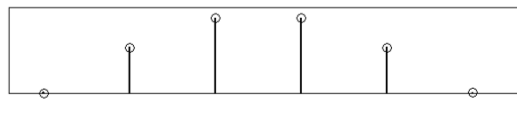
T/H signal  
(also a continuous-time signal)



clock



discrete-time signal



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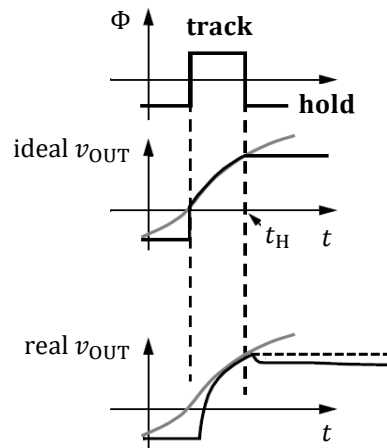
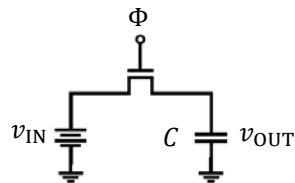
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## Real Track & Hold



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  - $kT/C$  Noise
  - Aperture Uncertainty
  - Signal Dependent Sampling Instant
  - Feedthrough and Drop (Hold Mode)
  - Track Mode Nonlinearity ( $R = f(v_{IN})$ )
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## Finite Acquisition Time I

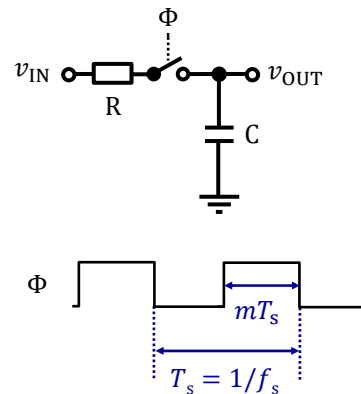
Finite speed in track mode due to time constant  $\tau = RC$  (lowpass filter)

What are the constraints on  $\tau = RC$  for a given sampling rate and resolution?

worst case

- switch open,  $v_{OUT} = 0$  V
- switch closes,  $v_{IN} = V_{FS}$
- $v_{OUT}$  must settle within fraction of LSB within time period  $mT_s$  (typically:  $m = 0.5$ )

simple model of a real switch  
= ideal switch + series resistance  $R$



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## Finite Acquisition Time II

$$v_{OUT} = V_{FS}(1 - e^{-t/\tau})$$

settle within  $\alpha LSB$

$$V_{FS} - \alpha LSB \leq V_{FS}(1 - e^{-t/\tau})$$

with  $V_{FS} = 2^B LSB$

$$\alpha LSB \geq 2^B LSB e^{-mT_s/\tau}$$

number of settling time constants  $M$

$$M = \frac{mT_s}{\tau} \geq \ln\left(\frac{2^B}{\alpha}\right)$$

| B  | M ( $\alpha = 0.5$ ) |
|----|----------------------|
| 6  | 4.9                  |
| 10 | 7.6                  |
| 14 | 10.4                 |
| 18 | 13.2                 |

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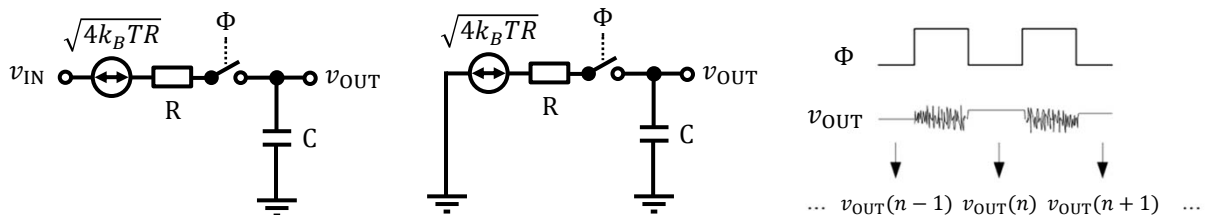
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## Thermal Noise



Obviously: A sampled noise value corresponds to the last instantaneous value of the noise at the end of a track period.

What is the spectrum of the discrete-time sequence representing these samples?

What is the power of the noise samples in hold mode?

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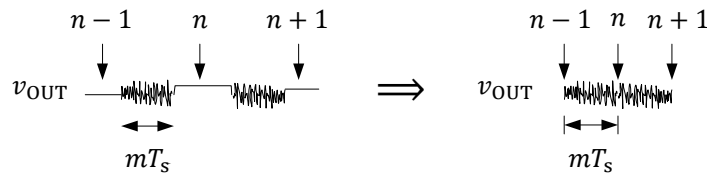
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## Spectrum of Noise Samples



Note that the noise process is interrupted in the hold mode.

⇒ Discrete-time noise samples are essentially  $mT_s$  apart instantaneous values of the continuous-time noise process in track mode.

Spectrum follows from the Fourier transform (step 2) of the process' autocorrelation function (step 1) (Wiener-Khintchin Theorem)  
 samples are uncorrelated → white spectrum  
 samples are correlated → colored spectrum

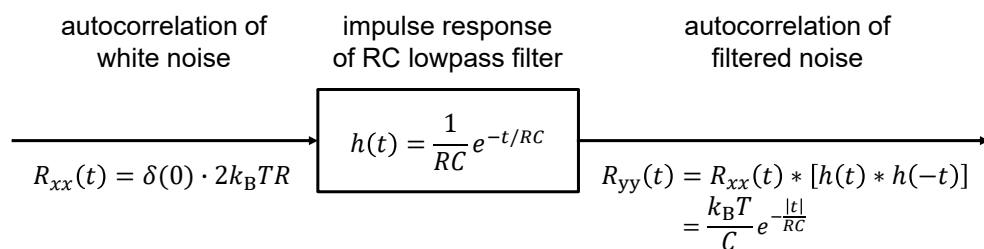
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## Step 1 – Autocorrelation Function



$$R_{yy}(n) = \frac{k_B T}{C} e^{-\frac{|n \cdot mT_s|}{RC}}$$

covariance of  $n$  samples  
separated by  $mT_s$

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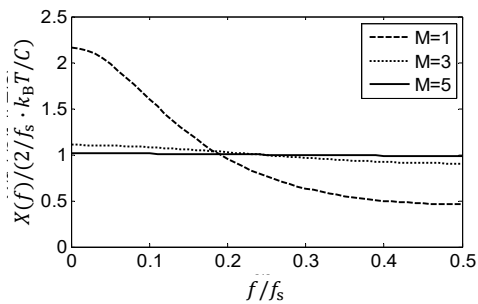
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## Step 2 – Fourier Transform

$$X(\omega) = \sum_{-\infty}^{\infty} R_{yy}(n) e^{-i\omega n T_s}$$

$$X(f) = \frac{2}{f_s} \frac{k_B T}{C} \frac{1 - e^{-2M}}{1 - 2e^{-M} \cos(2\pi \frac{f}{f_s}) + e^{-2M}} \quad \text{with } M = \frac{m T_s}{RC}$$



Spectrum is white for  $M > 3$ .

Makes sense: Large  $M$   
 $\Rightarrow$  Noise sample values decay from one cycle to the next.

Remember:  
 resolution  $B = 6 \Rightarrow M = 4.9$

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## Noise Power & Implications

### Parseval's theorem

The power of a time-domain signal is equal to its power spectral density integrated over all frequencies (see Analog CMOS Circuit Design).

$$\overline{v_{n,out}^2} = \int_0^{\infty} PSD_{n,out}(f) df = \int_0^{\infty} \left| \frac{1}{1 + sRC} \right|^2 4k_B T R df = \frac{k_B T}{C}$$

$\frac{k_B T}{C}$  - noise typically limits the resolution of an A/D converter  
 $\rightarrow$  make it equal to the quantization noise

$$\frac{k_B T}{C} = \frac{\Delta^2}{12} \Rightarrow C = 12k_B T \left( \frac{2^B}{V_{FS}} \right)^2$$

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Example –  $V_{FS} = 1\text{ V}$ ,  $\alpha = 0.5$ ,  $m = 0.5$ ,  $f_s = 100\text{ MHz}$

$$\frac{k_B T}{C} = \frac{\Delta^2}{12} \Rightarrow C = 12 k_B T \left( \frac{2^B}{V_{FS}} \right)^2$$

number of settling time constants  $M$

$$M = \frac{m T_s}{\tau} = \frac{m T_s}{RC} \geq \ln \left( \frac{2^B}{\alpha} \right)$$

$$R \leq \frac{m}{C f_s \ln \left( \frac{2^B}{\alpha} \right)}$$

| $B$ | $C$ (pF)  | $R$ ( $\Omega$ ) |
|-----|-----------|------------------|
| 8   | 0.003     | 246,057.00       |
| 10  | 0.052     | 12,582.00        |
| 12  | 0.834     | 665.00           |
| 14  | 13.300    | 36.00            |
| 16  | 213.000   | 1.99             |
| 18  | 3,416.000 | 0.11             |

Oversampling allows to reduce the capacitor size  
(will be discussed later).

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Example – LT2242-12 (12-bit 250 MSps Pipeline ADC)

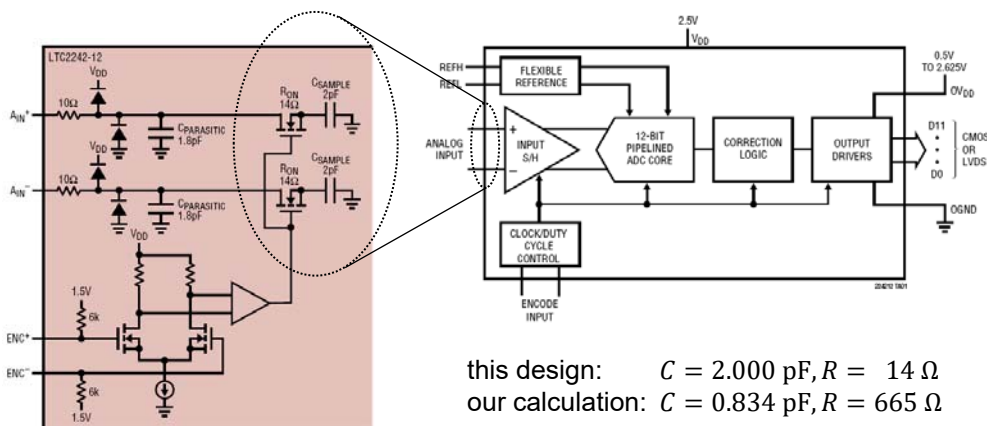


Figure 2. Equivalent Input Circuit

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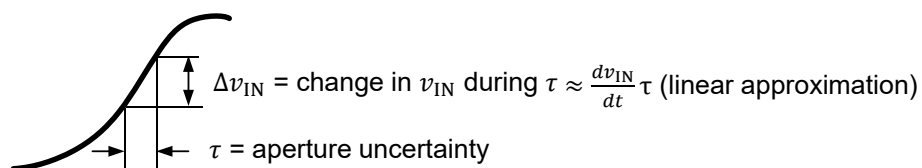
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## Aperture Uncertainty

Any sampling circuit shows random timing variations in the actual sampling clock edge.



result: sampling error ("noise"), particularly if  $dv_{IN}/dt$  is large.

analysis

- consider sine wave input signal
- assume  $\tau$  is random with zero mean and standard deviation  $\sigma_t$

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## Analysis

$$\Delta v_{IN} \approx \frac{dv_{IN}}{dt} \tau$$

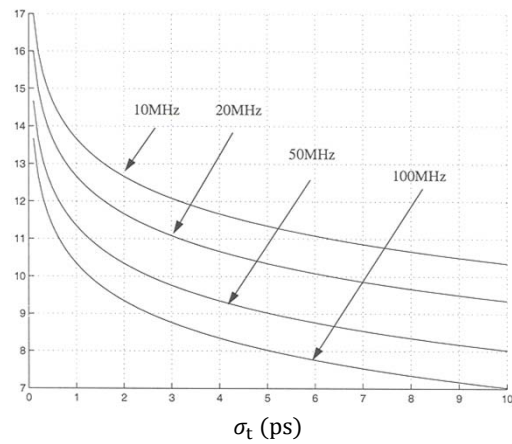
$$E[\Delta v_{IN}^2] = E\left[\left(\frac{dv_{IN}}{dt}\right)^2 \tau^2\right] = E\left[\left(\frac{dv_{IN}}{dt}\right)^2\right] E[\tau^2]$$

$$= E\left[\left(\frac{d}{dt} \hat{u} \sin(2\pi f t)\right)^2\right] \sigma_t^2$$

$$\approx \frac{1}{2} (\hat{u} 2\pi f)^2 \sigma_t^2$$

$$SNR_{aperture} \approx 10 \log \frac{1}{(2\pi f \sigma_t)^2}$$

ENOB (bits)



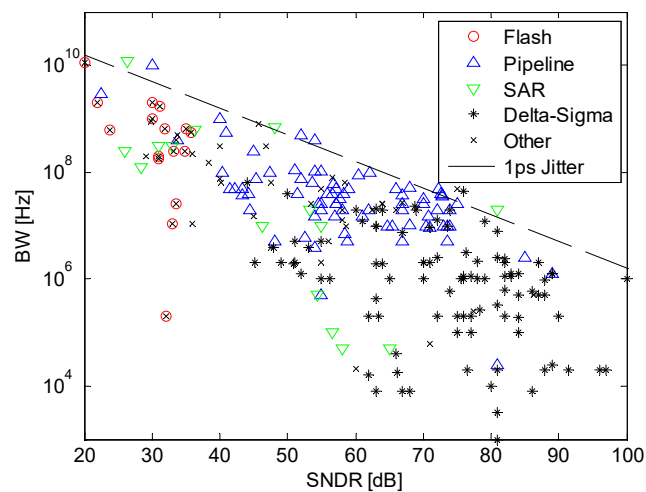
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## ADC Performance Survey (ISSCC & VLSI 97-23)



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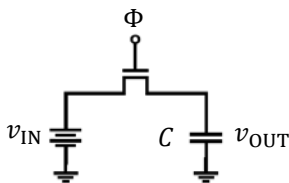
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## Voltage Dependence of Switch



$$i_D = \beta_0 \frac{W}{L} \left( v_{GS} - V_{th} - \frac{v_{DS}}{2} \right) v_{DS}$$

$$R_{on} \approx \left( \frac{di_D}{dv_{DS}} \Big|_{v_{DS} \rightarrow 0 V} \right)^{-1} = \frac{1}{\beta_0 \frac{W}{L} (v_{GS} - V_{th})}$$

$$R_{on} = \frac{1}{\beta_0 \frac{W}{L} (\Phi - v_{IN}(t) - V_{th})}$$

two problems:

- **transistor's turn-off is signal dependent ( $\Phi = v_{IN} + V_{th}$ )**
- $R_{on}$  is modulated by  $v_{IN}(t)$  (for  $\Phi = V_{DD} = \text{constant}$ ), i.e.,  $R_{on} = R_{on}(t)$   
(discussed in subchapter *Track Mode Nonlinearity*)

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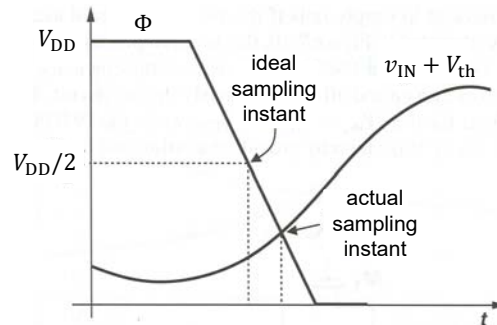
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## Signal Dependent Sampling Instant



solution

- make fall time of sampling clock much faster than maximum  $dv_{IN}/dt$  so that the sampling error is limited to a fraction of an LSB
  - typically not a show-stopper

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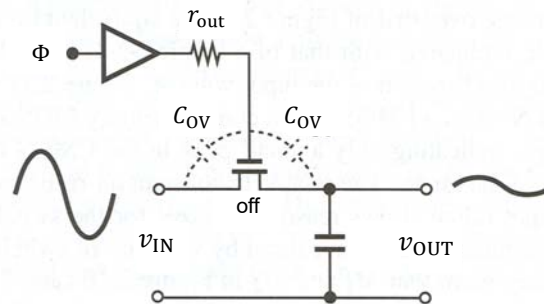
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## Signal Feedthrough in Hold Mode



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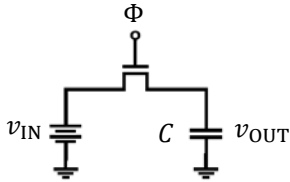
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## Voltage Dependence of Switch



$$i_D = \beta_0 \frac{W}{L} \left( v_{GS} - V_{th} - \frac{v_{DS}}{2} \right) v_{DS}$$

$$R_{on} \approx \left( \frac{di_D}{dv_{DS}} \bigg|_{v_{DS} \rightarrow 0} \right)^{-1} = \frac{1}{\beta_0 \frac{W}{L} (v_{GS} - V_{th})}$$

$$R_{on} = \frac{1}{\beta_0 \frac{W}{L} (\Phi - v_{IN}(t) - V_{th})}$$

two problems:

- transistor's turn-off is signal dependent ( $\Phi = v_{IN} + V_{th}$ )
- **$R_{on}$  is modulated by  $v_{IN}(t)$  (for  $\Phi = V_{DD} = \text{constant}$ ), i.e.,  $R_{on} = R_{on}(t)$**

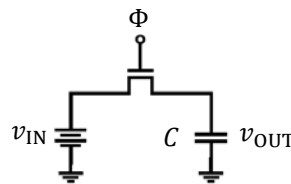
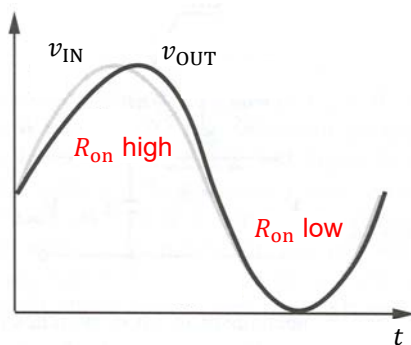
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## Track Mode Nonlinearity



$$R_{on} = \frac{1}{\beta_0 \frac{W}{L} (\Phi - v_{IN}(t) - V_{th})}$$

- output tracks well if input voltage is low
- output is distorted when input voltage is high due to increased  $R_{on}$

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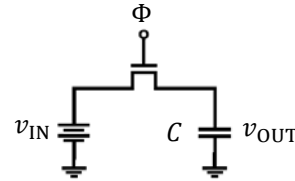
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## Analysis

$$C \frac{dv_{OUT}}{dt} = i_D = \beta_0 \frac{W}{L} \left( v_{GS} - V_{th} - \frac{v_{DS}}{2} \right) v_{DS}$$

$$\Leftrightarrow C \frac{dv_{OUT}}{dt} = \beta_0 \frac{W}{L} (\Phi - v_{IN} - V_{th})(v_{OUT} - v_{IN}) - \frac{\beta_0}{2} (v_{OUT} - v_{IN})^2$$



- “all” we need to do is to solve this non-linear differential equation...
- alternative: apply Volterra series analysis
  - approach to the calculation of the frequency domain response of nonlinear circuits with memory
- someone has already done this:  
W. Yu et al., „Distortion Analysis of MOS Track-and-Hold Sampling Mixers Using Time-Varying Volterra Series,” IEEE Trans. Circ. Syst. II, pp. 101-113, Feb. 1999

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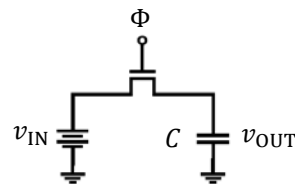
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## Result

$$|HD_3| = \frac{\text{amplitude of third harmonic}}{\text{amplitude of fundamental}}$$

$$\approx \frac{1}{4} \frac{\hat{u}^2}{(V_{GS} - V_{th})^2} \frac{2\pi f C}{\beta (V_{GS} - V_{th})}$$

$$\approx \frac{1}{4} \frac{\hat{u}^2}{(V_{GS} - V_{th})^2} 2\pi f C R_{ON} = \frac{1}{4} \frac{\hat{u}^2}{(V_{GS} - V_{th})^2} \frac{\omega}{\omega_c}$$



- $V_{GS}$  and  $R_{ON}$  are the bias / “quiescent” point values
- for low distortion
  - make amplitude  $\hat{u}$  smaller than  $V_{GS} - V_{th} \rightarrow$  low signal swing
  - make corner frequency  $\omega_c$  much larger than  $\omega \rightarrow$  big switch

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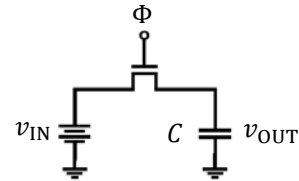
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## Example

$$|HD_3| \approx \frac{1}{4} \frac{\hat{u}^2}{(V_{GS} - V_{th})^2} 2\pi f C R_{ON} = \frac{1}{4} \frac{\hat{u}^2}{(V_{GS} - V_{th})^2} \frac{\omega}{\omega_c}$$



- signal is biased at  $V_{DD}/2 = 0.9 \text{ V}$ , signal frequency  $f = f_s/2$ , signal amplitude  $\hat{u} = 0.2 \text{ V}$
- $V_{GS} - V_{th} = 1.8 \text{ V} - 0.9 \text{ V} - 0.45 \text{ V} = 0.45 \text{ V}$
- $\omega/\omega_c = 1/20$

$$|HD_3| \approx \frac{1}{4} \frac{0.2^2}{0.45^2} \frac{1}{20} \rightarrow -52 \text{ dB}$$

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## Measured Data

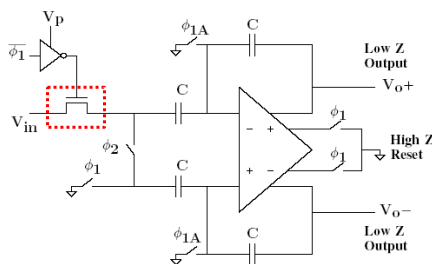
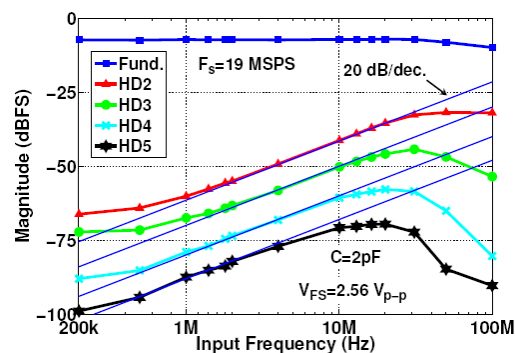


Fig. 3. S/H test circuit.

Fig. 5. Measured HD vs.  $f_{in}$ ,  $F_s = 19 \text{ MSPS}$ .

Brown et al., "Prediction and Characterization of Frequency Dependent MOS Switch Linearity and the Design Implications," CICC 2006

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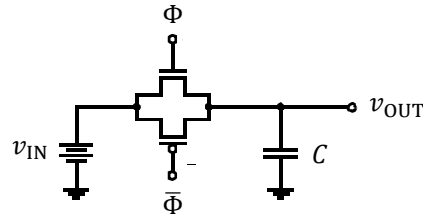
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## Compensation Approach I – Transmission Gate

$$\begin{aligned}
 R_{on} &= \frac{1}{\beta_{on} \frac{W}{L} \Big|_n (V_{GSn} - V_{thn})} \parallel \frac{1}{\beta_{op} \frac{W}{L} \Big|_p (V_{GSp} - |V_{thp}|)} \\
 &= \frac{1}{\beta_{on} \frac{W}{L} \Big|_n (V_{DD} - v_{IN} - V_{thn})} \parallel \frac{1}{\beta_{op} \frac{W}{L} \Big|_p (|v_{IN}| - |V_{thp}|)} \\
 &= \frac{1}{\beta_{on} \frac{W}{L} \Big|_n (V_{DD} - V_{thn} - |V_{thp}|)} \quad \text{if } \beta_{on} \frac{W}{L} \Big|_n = \beta_{op} \frac{W}{L} \Big|_p
 \end{aligned}$$



- seems to be independent of  $v_{IN}$
- However, the requirement  $\beta_{on} \frac{W}{L} \Big|_n = \beta_{op} \frac{W}{L} \Big|_p$  can never be met in practice.
- Moreover, back-gate and short-channel effects are not considered!

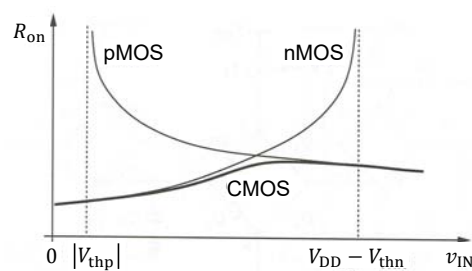
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## Real Transmission Gate



- Size p/n device ratio to minimize change in  $R_{on}$  over desired input range.
- Size pMOS and nMOS simultaneously to meet distortion specifications.
- remaining issue: pMOS and nMOS turn off at slightly different times

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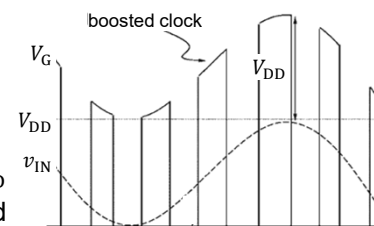
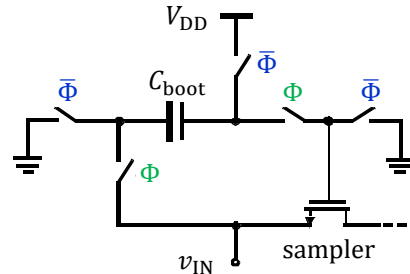
## Compensation Approach II – Bootstrapping

phase  $\bar{\Phi}$

- $C_{boot}$  is pre-charged to  $V_{DD}$
- sampling switch is off

phase  $\Phi$

- $C_{boot}$  is connected as a battery between gate and source of the sampling switch  $\rightarrow$  constant  $V_{GS}$
- moves with the input voltage but  $V_{GS} = V_{DD}$  is constant



Note that  $V_{GB}$  may become as large as  $2V_{DD}$   
 $\rightarrow$  oxide breakdown must be prevented

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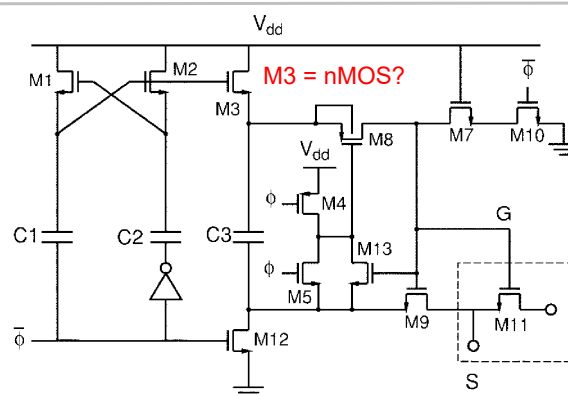
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## Example – Implementation

voltage doubler to turn  
M3 (nMOS) on and  
load C3 to  $V_{DD}$



typically  
 $V_{Gmax} = 1.3 V_{DD}$

sampler

A. Abo et al, "A 1.5-V, 10-bit, 14.3-MS/s CMOS Pipeline Analog-to-Digital Converter,"  
 IEEE J. Solid-State Circ., pp. 599, May 1999

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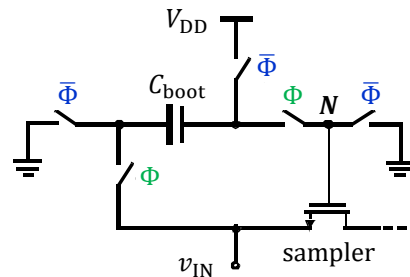
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## Limitations

- Ideally, switch on-resistance  $R_{on}$  is independent of input signal.
- In practice, parasitic capacitors at gate node  $N$  and the body effect limit the achievable linearity.



$$R_{on} = \frac{1}{\beta_0 \frac{W}{L} (v_{GS} - V_{th})}$$

$$= \frac{1}{\beta_{0n} \frac{W}{L} \left( \frac{C_{boot}}{C_{boot} + C_N} V_{DD} - \frac{C_N}{C_{boot} + C_N} v_{IN} - V_{th}[v_{IN}] \right)}$$

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## Alternative Implementation

- simpler implementation of pre-charging and switching
- higher parasitic capacitor at the gate of the switching transistor

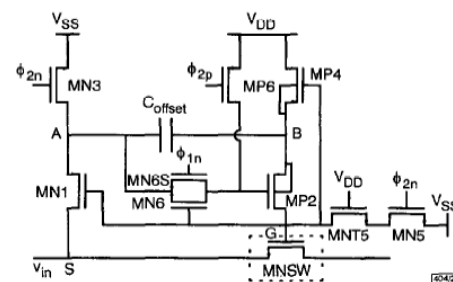


Fig. 2 Proposed implementation

Dessouky and Kaiser, "Input Switch Configuration Suitable for Rail-to-Rail Operation of Switched Opamp Circuits," Electronic Letters, January 1999

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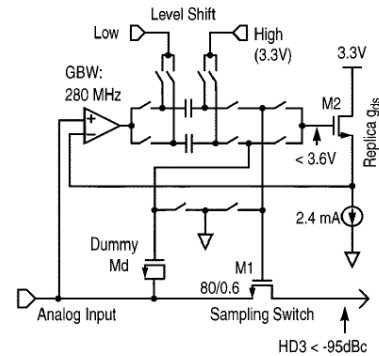


## Advanced Bootstrapping Schemes I

- an attempt to cancel back-gate effect of switch transistor M1 by means of replica transistors M2
- source-bulk voltage  $v_{SB}$  is not constant  
→ modulates threshold voltage  $V_{th}$  and thus  $R_{on}$

$$R_{on} \approx \left( \frac{di_D}{dv_{DS}} \bigg|_{V_{DS} \rightarrow 0V} \right)^{-1} = \frac{1}{\beta_0 \frac{W}{L} (v_{GS} - V_{th})}$$

- amplifier required  
→ its bandwidth limits the speed of the sampler



H. Pan et al., "A 3.3-V 12-b 50-MS/s A/D Converter in 0.6μm CMOS with over 80-dB SFDR,"  
*IEEE J. Solid-State Circ.*, pp. 1769-1780, Dec. 2000

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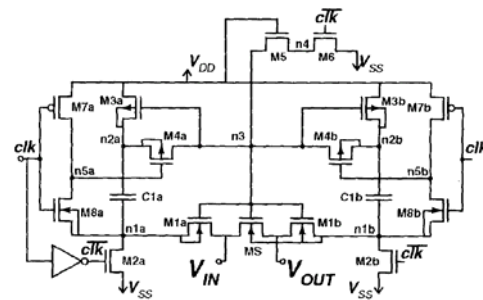
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## Advanced Bootstrapping Schemes II

- Which side of the sampling switch is source, which one is drain?
- gate tracks average of input and output  
→ reduced distortion due to voltage drop over  $R_{on}$
- bulk tracks signal  
→ reduced back-gate effect
- measured  $SFDR = 76.5$  dB at  $f_{sig} = 200$  MHz



M. Waltari et al., "A Self-Calibrated Pipeline ADC with 200MHz IF-Sampling Frontend,"  
*ISSCC 2002, Dig. Techn. Papers*, pp. 314

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## Overview

- Introduction
- **Nonidealities**
  - Finite Acquisition Time
  - $kT/C$  Noise
  - Aperture Uncertainty
  - Signal Dependent Sampling Instant
  - Feedthrough and Drop (Hold Mode)
  - Track Mode Nonlinearity ( $R = f(v_{IN})$ )
  - **Pedestal Error / Charge Injection**
- Bottom Plate Sampling
- Design Considerations

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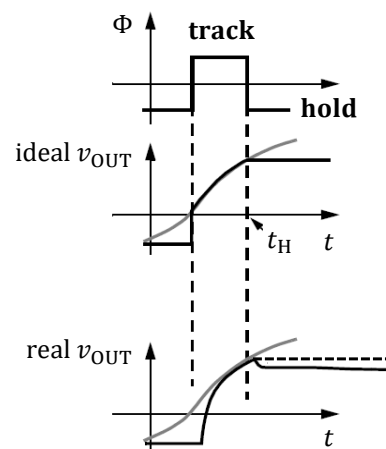
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## Pedestal Error

- error introduced at the output during transition from track to hold
- caused by charge injection (= charge from overlap capacitors & channel)
- depends on the duration of the clock transition (= waveform of  $\Phi$ )
- two extrema
  - „quasi-static gating“ / „slow gating“  
→ charge from overlap capacitors
  - „fast gating“  
→ charge from channel



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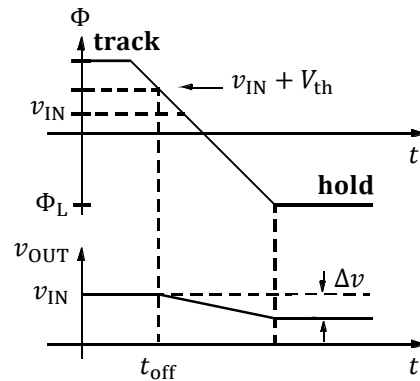
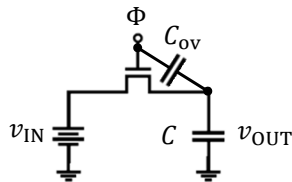
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## Slow Gating

- $t_{\text{off}} \rightarrow \Phi = v_{\text{IN}} + V_{\text{th}}$   
(see "Signal Dependent Sampling Instant")
- Channel charge has disappeared by  $t_{\text{off}}$  without introducing an error.
- However, there still is charge stored in the overlap capacitor that will be redistributed until  $\Phi$  reaches  $\Phi_L \rightarrow$  apply principle of charge conservation to determine  $\Delta v$



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## Slow Gating – Derivation

@  $t_{\text{off}}$ :  $Q_1 = -C_{\text{ov}}V_{\text{th}} + C v_{\text{IN}}$

hold phase:  $Q_2 = C_{\text{ov}}[(v_{\text{IN}} - \Delta v) - \Phi_L] + C(v_{\text{IN}} - \Delta v)$

Solving  $Q_1 = Q_2$  yields

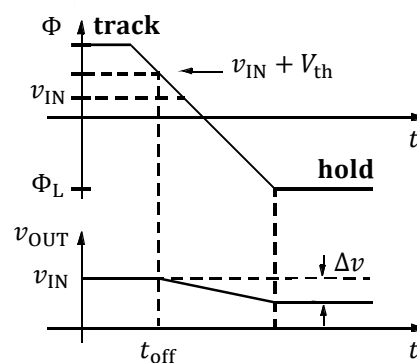
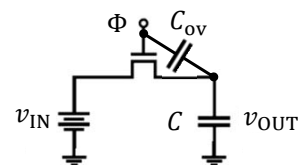
$$\Delta v = \frac{C_{\text{ov}}}{C + C_{\text{ov}}} v_{\text{IN}} + \frac{C_{\text{ov}}}{C + C_{\text{ov}}} (V_{\text{th}} - \Phi_L)$$

From  $v_{\text{OUT}} = v_{\text{IN}} - \Delta v$ , it follows

$$v_{\text{OUT}} = (1 + \varepsilon) v_{\text{IN}} + V_{\text{OS}}$$

with

$$\varepsilon = -\frac{C_{\text{ov}}}{C + C_{\text{ov}}} \quad \text{and} \quad V_{\text{OS}} = -\frac{C_{\text{ov}}}{C + C_{\text{ov}}} (V_{\text{th}} - \Phi_L)$$



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## Slow Gating – Example

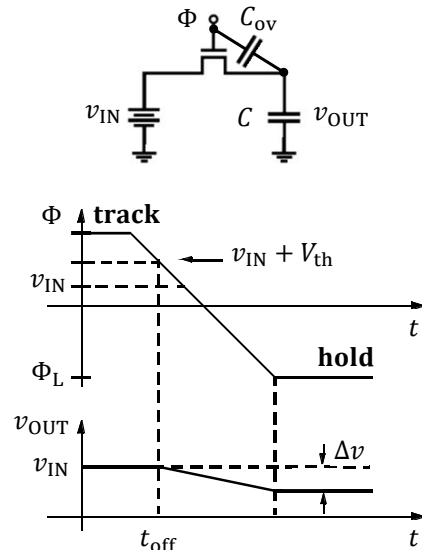
$$v_{OUT} = (1 + \varepsilon)v_{IN} + V_{OS}$$

$$\varepsilon = -\frac{C_{ov}}{C + C_{ov}} \quad \text{with} \quad V_{OS} = -\frac{C_{ov}}{C + C_{ov}}(V_{th} - \Phi_L)$$

For

- $C = 1 \text{ pF}$
- $\Phi_L = 0 \text{ V}, V_{th} = 250 \text{ mV}$
- $W = 20 \text{ }\mu\text{m}, C_{GD0} = 0.1 \frac{\text{fF}}{\mu\text{m}} \rightarrow C_{ov} = 2 \text{ fF}$

$$v_{OUT} = (1 - 0.2 \%)v_{IN} - 0.5 \text{ mV}$$



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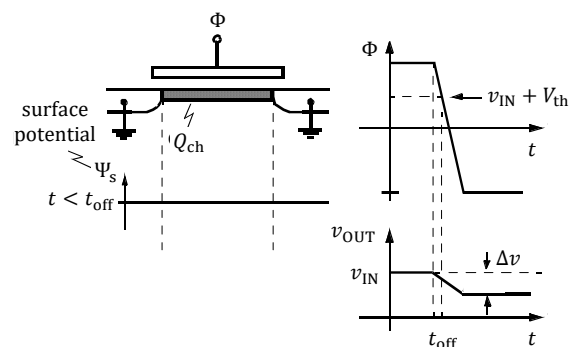
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## Fast Gating

- The channel charge cannot change instantaneously.
- The resulting surface potential decays due to a charge flow to source and drain.
- The charge distribution is determined by the impedances at source and drain.



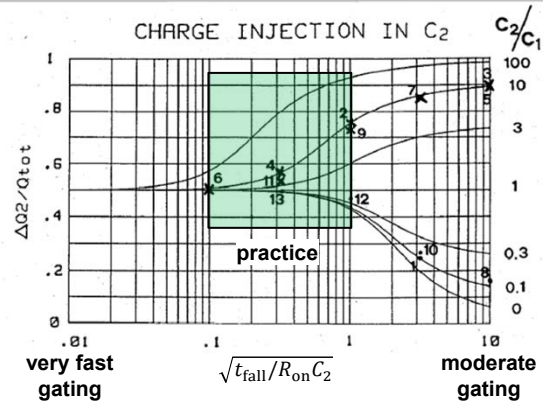
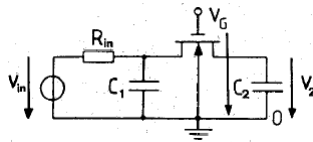
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## Charge Distribution



G. Wegmann et al., "Charge Injection in Analog MOS Switches,"  
IEEE J. Solid-State Circ., pp. 1091-1097, June 1987

Y. Ding and R. Harjani, "A Universal Analytic Charge Injection Model,"  
Proc. ISCAS, pp. 144-147, 2000

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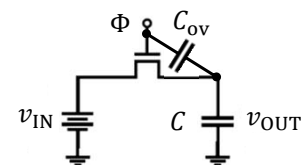
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## Fast Gating – Derivation

### Assumptions

- transistor immediately turns off when  $\Phi$  switches from  $\Phi_H$  to  $\Phi_L$
- charge splits equally between source and drain



$$\Delta v = \frac{C_{ov}}{C + C_{ov}} v_{IN} + \frac{C_{ov}}{C + C_{ov}} (V_{th} - \Phi_L) \text{ slow gating}$$

$$\Delta v = \frac{C_{ov}}{C + C_{ov}} (\Phi_H - \Phi_L) + \frac{1}{2} \frac{Q_{ch}}{C} \text{ with } Q_{ch} = C'_{ox} WL (\Phi_H - v_{IN} - V_{th})$$

$$v_{OUT} = (1 + \varepsilon) v_{IN} + V_{OS}$$

$$\text{with}$$

$$\varepsilon = -\frac{1}{2} \frac{C'_{ox} WL}{C} \text{ and } V_{OS} = -\frac{C_{ov}}{C + C_{ov}} (\Phi_H - \Phi_L) - \frac{1}{2} \frac{C'_{ox} WL}{C} (\Phi_H - V_{th})$$

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## Fast Gating – Example

$$v_{\text{OUT}} = (1 + \varepsilon)v_{\text{IN}} + V_{\text{OS}}$$

with

$$\varepsilon = -\frac{1}{2} \frac{C'_{\text{ox}} W L}{C} \quad \text{and} \quad V_{\text{OS}} = -\frac{C_{\text{ov}}}{C + C_{\text{ov}}} (\Phi_{\text{H}} - \Phi_{\text{L}}) - \frac{1}{2} \frac{C'_{\text{ox}} W L}{C} (\Phi_{\text{H}} - V_{\text{th}})$$

For

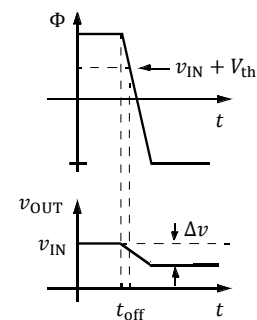
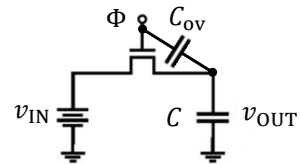
- $C = 1 \text{ pF}$
- $\Phi_{\text{H}} = 1.8 \text{ V}, \Phi_{\text{L}} = 0 \text{ V}, V_{\text{th}} = 250 \text{ mV}$
- $W = 20 \text{ } \mu\text{m}, C_{\text{GD0}} = 0.1 \frac{\text{fF}}{\mu\text{m}} \rightarrow C_{\text{ov}} = 2 \text{ fF}$
- $C'_{\text{ox}} W L = 40 \text{ fF}$

$$v_{\text{OUT}} = (1 - 2.0 \%) v_{\text{IN}} - 35 \text{ mV}$$

fast gating

$$v_{\text{OUT}} = (1 - 0.2 \%) v_{\text{IN}} - 0.5 \text{ mV}$$

slow gating



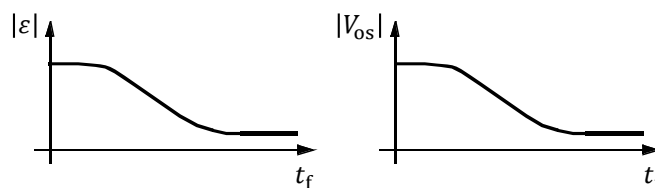
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## Transition Fast / Slow Gating



$|\varepsilon|$  and  $|V_{\text{OS}}|$  decrease as the fall time  $t_f$  of  $\phi$  increases  
and  
approach the limit case of slow gating.

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## Speed-Accuracy Trade-off

$$\Delta v \approx \frac{1}{2} \frac{Q_{ch}}{C}$$

$$Q_{ch} = C'_{ox} W L (v_{GS} - V_{th})$$

$$R_{on} = \frac{1}{\beta_0 \frac{W}{L} (v_{GS} - V_{th})} = \frac{1}{\mu C'_{ox} \frac{W}{L} (v_{GS} - V_{th})} = \frac{L^2}{\mu Q_{ch}} \rightarrow Q_{ch} = \frac{L^2}{\mu R_{on}}$$

$$M = \frac{mT_s}{\tau} = \frac{1}{2f_s R_{on} C} \rightarrow f_s = \frac{1}{2MR_{on}C}$$

$$\frac{\Delta v}{f_s} \approx M \frac{L^2}{\mu}$$

example

- $L = 0.12 \mu\text{m}$ ,  $\mu = 415 \text{ cm}^2/\text{Vs}$  (Analog CMOS)
- $M = 10.4$  (14 bits) and  $M = 13.2$  (18 bits)

$$3.6 \frac{\text{mV}}{\text{GHz}} \leq \frac{\Delta v}{f_s} \leq 4.6 \frac{\text{mV}}{\text{GHz}}$$

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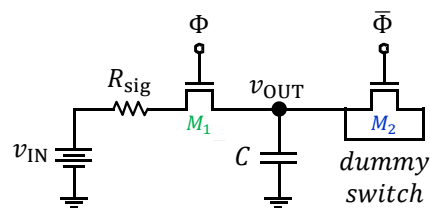
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## Compensation Approach I – Charge Cancellation

$M_1$  turns off &  $M_2$  turns on  
 $\rightarrow$  charge delivered by  $M_1$  is  
 absorbed/compensated by  $M_2$

$$\Delta Q_1 \approx 0.5 Q_{ch1} + Q_{ov1} = \Delta Q_2 \approx Q_{ch2} + 2Q_{ov2}$$

$$\rightarrow W_2 = 0.5W_1 \text{ if } L_2 = L_1$$



The cancellation is never perfect since the channel  
 charge of  $M_1$  will not exactly split 50/50.

C. Eichenberger and W. Guggenbuehl, "Dummy Transistor Compensation of Analog MOS Switches,"  
*IEEE J. Solid-State Circ.*, vol. 24, no. 4, pp. 1143-1146, Aug. 1989

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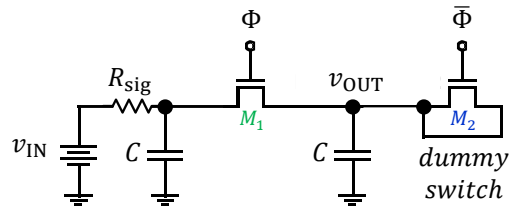
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## Compensation Approach II – Equalization Capacitor



Much better cancellation since impedances at source and drain are “equalized”.  
drawback: capacitor at input reduces the bandwidth

L. Bienstman and H. De Man, “An Eight-Channel 8 Bit Microprocessor Compatible NMOS D/A Converter with Programmable Scaling,”  
*IEEE J. Solid-State Circ.*, vol. 15, no. 6, pp. 1051-1059, Aug. 1980

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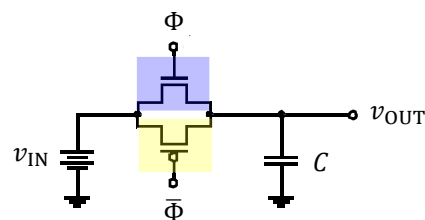
## Compensation Approach III – Transmission Gate

$$Q_{ch,n} = C'_{ox,n} W_n L_n (v_{GS} - V_{th,n})$$

$$= C'_{ox,n} W_n L_n (\Phi_H - v_{IN} - V_{th,n})$$

$$Q_{ch,p} = C'_{ox,p} W_p L_p (|v_{GS}| - |V_{th,p}|)$$

$$= C'_{ox,p} W_p L_p (v_{IN} - \Phi_L - |V_{th,p}|)$$



$$\text{fast gating: } \Delta v = \frac{C_{ov}}{C + C_{ov}} (\Phi_H - \Phi_L) + \frac{1}{2} \frac{Q_{ch}}{C} \approx \frac{1}{2} \frac{Q_{ch}}{C} = \frac{1}{2} \frac{Q_{ch,p} - Q_{ch,n}}{C}$$

- achieves partial cancellation due to signal dependency
- design approach for charge equalization:  $W_p L_p = W_n L_n$

versus

design approach for optimizing track mode linearity & speed:  $L_p = L_n = L_{min} \rightarrow W_p = 2 \dots 3 W_n$

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## Compensation Approach IV – Differential Sampling

$$v_{IN,DM} = v_{IN1} - v_{IN2} \quad v_{OUT,DM} = v_{OUT1} - v_{OUT2}$$

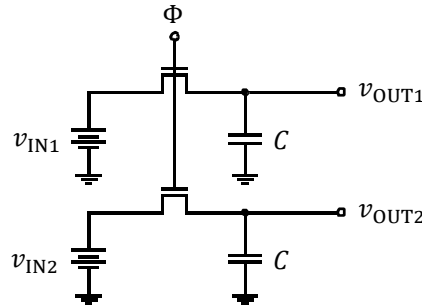
$$v_{IN,CM} = \frac{v_{IN1} + v_{IN2}}{2} \quad v_{OUT,CM} = \frac{v_{OUT1} + v_{OUT2}}{2}$$

$$v_{OUT1} = (1 + \varepsilon_1)v_{IN1} + V_{OS1}$$

$$v_{OUT2} = (1 + \varepsilon_2)v_{IN2} + V_{OS2}$$

$$v_{OUT,DM} = \left(1 + \frac{\varepsilon_1 + \varepsilon_2}{2}\right)v_{IN,DM} + (\varepsilon_1 - \varepsilon_2)v_{IN,CM} + (V_{OS1} - V_{OS2})$$

$$v_{OUT,CM} = \left(\frac{\varepsilon_1 - \varepsilon_2}{4}\right)v_{IN,DM} + \left(1 + \frac{\varepsilon_1 + \varepsilon_2}{2}\right)v_{IN,CM} + \left(\frac{V_{OS1} + V_{OS2}}{2}\right)$$



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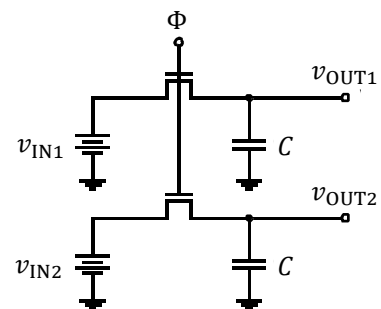
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## Differential Sampling – Characteristics

$$v_{OUT,DM} = \left(1 + \frac{\varepsilon_1 + \varepsilon_2}{2}\right)v_{IN,DM} + (\varepsilon_1 - \varepsilon_2)v_{IN,CM} + (V_{OS1} - V_{OS2})$$

- assuming good matching
  - only **small residual offset** in  $v_{OUT,DM}$
  - good rejection of coupling noise, substrate noise, ...
  - small common-mode to differential-mode crosstalk**
- However,  $v_{OUT,DM}$  still shows the **same gain error** as the single-ended implementation
- further issues
  - non-linear terms due to the body effect,...
  - even-order terms will cancel, but not uneven-order terms → limits  $HD_3$  and  $SFDR$



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## Overview

- Introduction
- Nonidealities
- **Bottom Plate Sampling**
  - **Introduction & Analysis**
  - Examples
- Design Considerations

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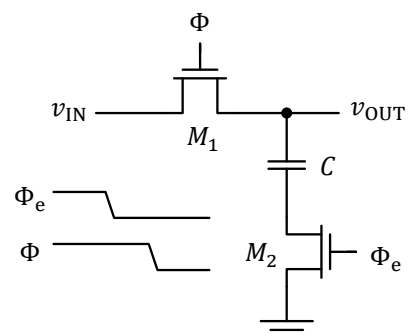
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## Introduction

- idea: sample the signal at the “grounded” plate of the capacitor to achieve signal independence
- developed in the 80's
  - D. J. Allstot and W. C. Black, Jr., “Technological Design Considerations for Monolithic MOS Switched-Capacitor Filtering Systems,” Proc. IEEE, pp. 967-986, Aug. 1983
  - K.-L. Lee and R. G. Meyer, “Low-Distortion Switched-Capacitor Filter Design Techniques,” IEEE J. Solid-State Circuits, pp. 1103-1113, Dec. 1985



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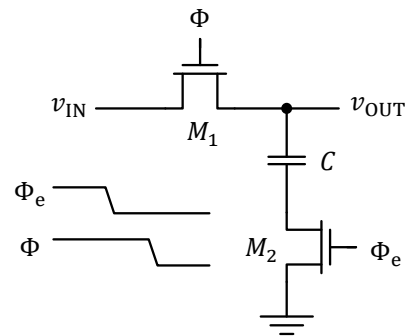
## Mode of Operation I

- $M_2$  turns off “slightly” before  $M_1$  (typically a few hundred picoseconds)
- charge  $\Delta Q_{\text{ch2}}$  injected by  $M_2$

$$\Delta Q_{\text{ch2}} \approx \alpha_2 C'_{\text{ox}} W_2 L_2 (\Phi_H - V_{\text{th}})$$

- to first order, charge  $\Delta Q_{\text{ch2}}$  injected by  $M_2$  is signal independent
- voltage  $V_C$  across  $C$

$$V_C = v_{\text{IN}} + \frac{\Delta Q_{\text{ch2}}}{C}$$



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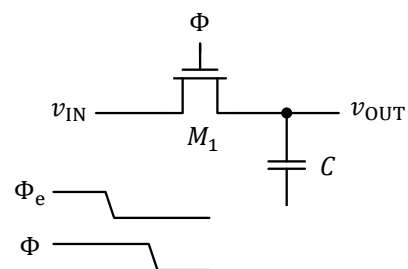
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## Mode of Operation II

- next,  $M_1$  turns off
- charge injected by  $M_1$

$$\Delta Q_{\text{ch1}} \approx \alpha_1 C'_{\text{ox}} W_1 L_1 (\Phi_H - v_{\text{IN}} - V_{\text{th}})$$

- However,  $C$  is floating  $\rightarrow$  no charge injection into  $C$
- Charge flows into signal source and/or parasitic capacitors at node  $v_{\text{OUT}}$ .
- To good to be true!  $\rightarrow$  There are parasitic capacitors at the bottom plate of  $C$ , e.g.,  $C_{\text{DB2}}$ .



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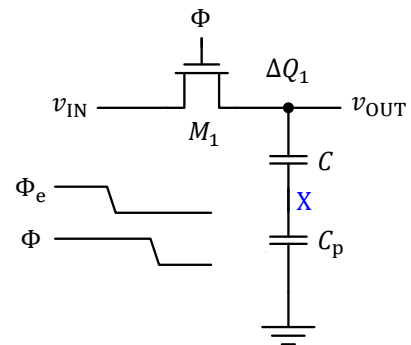
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## Mode of Operation III

- intuition: overall parasitic capacitor  $C_p$  is small and in series with  $C \rightarrow$  charge injection by  $M_1$  must be small
- Moreover, even if  $M_1$  injects charge into  $C$ , the total charge at node **X** can merely be redistributed but not change.
- idea: process charge at node **X** instead of voltage across sampling capacitor  $C$



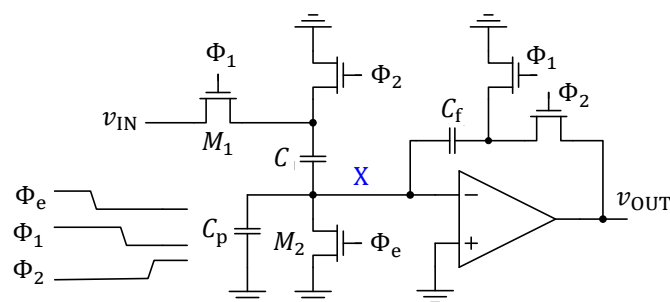
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## Charge Redistribution Track & Hold



Let's analyze the circuit by considering the different clock phases!

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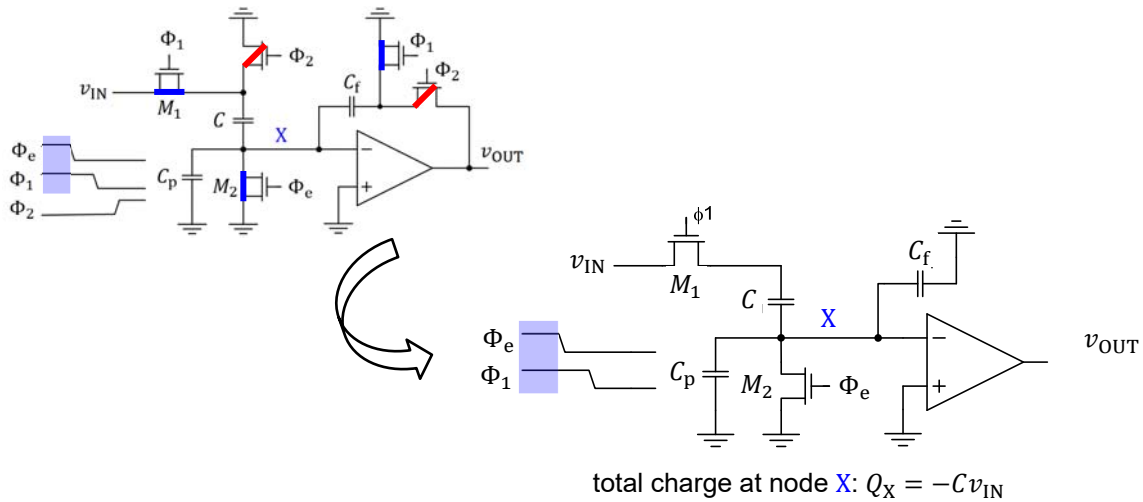
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## Phase I: $\Phi_1$ and $\Phi_e$ High



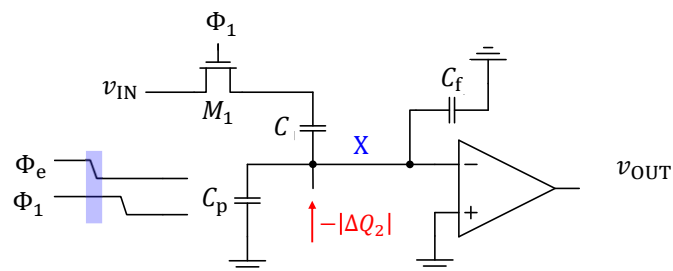
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## Phase II: $\Phi_1$ High, $\Phi_e$ Switching to Low



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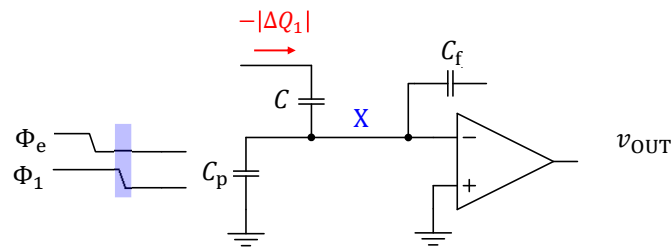
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### Phase III: $\Phi_1$ Switching to Low, $\Phi_e$ Low



Charge injection  $\Delta Q_1$  by  $M_1$  results in changes of all voltages across all capacitors.  
However, total charge at node  $X$  remains constant.

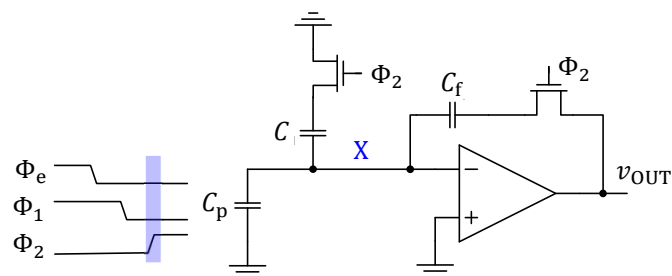
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### Phase IV: $\Phi_2$ Switching to High, $\Phi_1$ and $\Phi_e$ Low



Opamp forces voltage at node  $X$  to zero, whereby  $C$  and  $C_p$  are shorted and thus discharged  
→ total charge of node  $X$  accumulates at  $C_f$ :  $Q_X = -C_f v_{OUT}$ .

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## Analysis – Charge Conservation

sampled charge  $Q_X = -C v_{IN} - |\Delta Q_2|$

charge after redistribution  $Q_X = -C_f v_{OUT}$

charge conservation

$$-C v_{IN} - |\Delta Q_2| = -C_f v_{OUT} \rightarrow$$

$$v_{OUT} = \frac{C}{C_f} v_{IN} + \frac{|\Delta Q_2|}{C_f}$$

Output has signal independent offset.  
→ Go for a fully differential circuit!

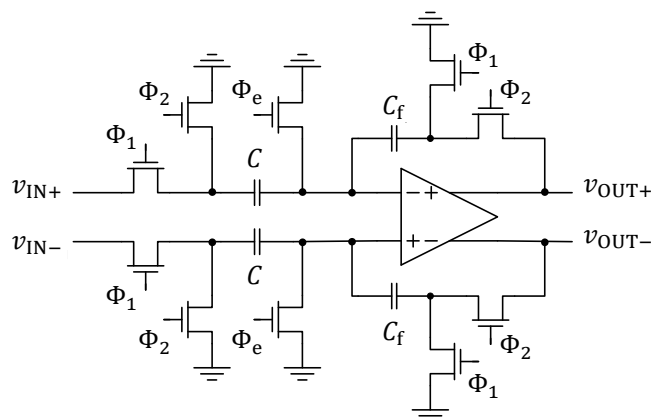
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## Fully Differential Charge Redistribution Track & Hold



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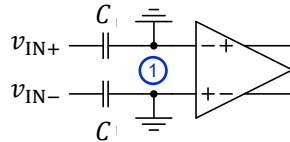
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## Analysis I – Charge Conservation

$\Phi_1$  and  $\Phi_e$  high



Once  $\Phi_e$  switched to low

$$\begin{aligned} Q_{1-} &= -C v_{IN+} - |\Delta Q| & Q_{2-} &= C v_- - C_f (v_{OUT+} - v_-) \\ Q_{1+} &= -C v_{IN-} - |\Delta Q| & Q_{2+} &= C v_+ - C_f (v_{OUT-} - v_+) \end{aligned}$$

$$\begin{aligned} Q_{1-} = Q_{2-} &\rightarrow v_{OUT+} = \frac{C}{C_f} v_{IN+} + \frac{|\Delta Q|}{C_f} + \frac{C + C_f}{C_f} v_- \\ Q_{1+} = Q_{2+} &\rightarrow v_{OUT-} = \frac{C}{C_f} v_{IN-} + \frac{|\Delta Q|}{C_f} + \frac{C + C_f}{C_f} v_+ \end{aligned}$$

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## Analysis II – Charge Conservation

$$v_{OUT+} - v_{OUT-} = \frac{C}{C_f} (v_{IN+} - v_{IN-}) \quad \text{since } v_- = v_+$$

No offset! However:

$$v_{OUT+} + v_{OUT-} = \frac{C}{C_f} (v_{IN+} + v_{IN-}) + 2 \frac{|\Delta Q|}{C_f} + \frac{C + C_f}{C_f} (v_+ + v_-)$$

$$v_{D,CM} = \frac{C_f}{C + C_f} v_{OUT,CM} - \frac{|\Delta Q|}{C + C_f} - \frac{C}{C + C_f} v_{IN,CM}$$

$$\text{since } v_{D,CM} = \frac{v_+ + v_-}{2}, v_{OUT,CM} = \frac{v_{OUT+} + v_{OUT-}}{2}, v_{IN,CM} = \frac{v_{IN+} + v_{IN-}}{2}$$

Common-mode input signals  $v_{IN,CM}$  result in common-mode signals  $v_{D,CM}$  at the amplifier's inputs.  
→ Amplifier with high common-mode rejection ratio (CMRR) required!

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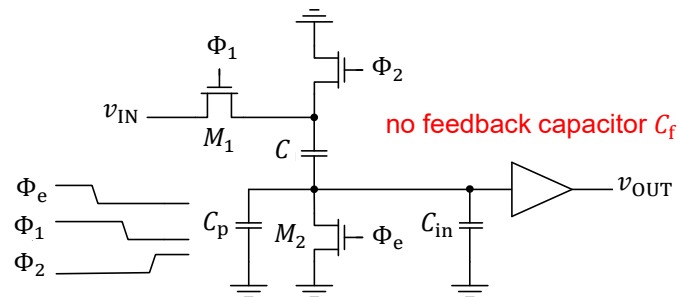
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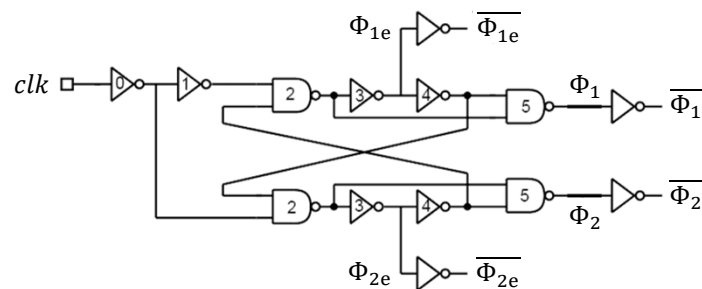
## Open-Loop T&H with Bottom-Plate Sampling



Signal dependent charge injection errors are reduced as in the charge-redistribution architecture. However, without a feedback capacitor  $C_f$ ,  $C$  forms a capacitive divider with  $C_p || C_{in}$  → introduces distortion, since  $C_p$  and  $C_{in}$  are typically weakly non-linear.



## Clock Generation – Example



Use a break-before-make circuit for generating  $\Phi_{1e}$ ,  $\Phi_1$  and  $\Phi_2$ .

A. Abo, "Design for Reliability of Low-voltage, Switched-capacitor Circuits,"  
PhD Thesis, UC Berkeley, 1999



## Overview

- Introduction
- Nonidealities
- **Bottom Plate Sampling**
  - Introduction & Analysis
  - **Examples**
- Design Considerations

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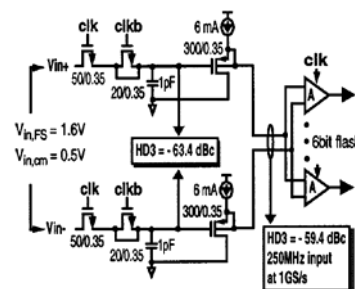
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## Low Precision High Speed T&H

A “simple” T&H circuit may be fine for low resolution, high speed applications.

- no bottom plate sampling
- just an optimized dummy transistor for cancellation of charge injection
- pMOS source-follower to buffer the sampled signal



M. Choi and A. Abidi, "A 6-b 1.3-Gsample/s A/D converter in 0.35- $\mu$ m CMOS,"  
*IEEE J. Solid-State Circuits*, pp.1847-1858, Dec. 2001

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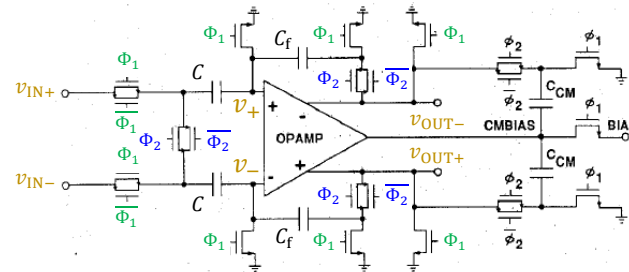
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## T&H with Common Mode Cancellation



- Shorting switch allows re-distributing differential charge on sampling capacitors  $C$ .
- Common mode at opamp inputs  $v_+$  and  $v_-$  gets independent of common mode at circuit's input  $v_{IN+}$  and  $v_{IN-}$ .

S.H. Lewis & P.R. Gray, "A Pipelined 5 MSample/s 9-bit Analog-to-Digital Converter",  
IEEE J. Solid-State Circuits, pp. 954-961, Dec. 1987

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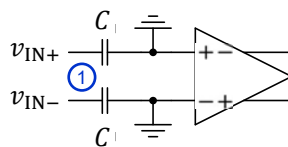
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## Analysis I – Charge Conservation @ Circuit's Input

$\Phi_1$  high



$$Q_{n1} = C v_{IN+} + C v_{IN-} \quad Q_{n1} = C(v_{\text{FLOAT}} - v_+) + C(v_{\text{FLOAT}} - v_-)$$

$$v_{\text{FLOAT}} = V_{\text{IN,CM}} + V_{\text{D,CM}}$$

$$\text{since } V_{\text{IN,CM}} = \frac{v_{\text{IN+}} + v_{\text{IN-}}}{2}, V_{\text{D,CM}} = \frac{v_+ + v_-}{2}$$

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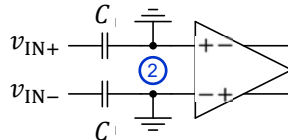
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## Analysis II – Charge Conservation @ Opamp's Inputs

$\Phi_1$  high



$$\begin{aligned} Q_{n2+} &= -C v_{IN+} \\ Q_{n2-} &= -C v_{IN-} \end{aligned}$$

$$\begin{aligned} Q_{n2+} &= -C(v_{FLOAT} - v_+) - C_f(v_{OUT-} - v_+) \\ Q_{n2-} &= -C(v_{FLOAT} - v_-) - C_f(v_{OUT+} - v_-) \end{aligned}$$

adding the charges yields

$$-C(v_{IN+} + v_{IN-}) = -2C v_{FLOAT} + C(v_+ + v_-) + C_f(v_+ + v_-) - C_f(v_{OUT+} + v_{OUT-})$$

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## Analysis II – Charge Conservation @ Opamp's Inputs

$$-C(v_{IN+} + v_{IN-}) = -2C v_{FLOAT} + C(v_+ + v_-) + C_f(v_+ + v_-) - C_f(v_{OUT+} + v_{OUT-})$$

$$\text{since } V_{IN,CM} = \frac{v_{IN+} + v_{IN-}}{2}, V_{D,CM} = \frac{v_+ + v_-}{2}, V_{OUT,CM} = \frac{v_{OUT+} + v_{OUT-}}{2}$$

$$-2C V_{IN,CM} = -2C v_{FLOAT} + 2C V_{D,CM} + 2C_f V_{D,CM} - 2C_f V_{OUT,CM}$$

Replacing  $v_{FLOAT} = V_{IN,CM} + V_{D,CM}$  (see slide Analysis I)

$$-2C V_{IN,CM} = -2C V_{IN,CM} - 2C V_{D,CM} + 2C V_{D,CM} + 2C_f V_{D,CM} - 2C_f V_{OUT,CM}$$

$$V_{D,CM} = V_{OUT,CM}$$

Amplifier's input common mode determined by output common mode, which is well regulated.

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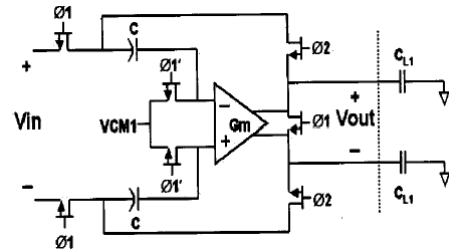
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## Flip-Around T&H

- Sampling capacitors are “flipped around” OTA and used as feedback capacitors during  $\Phi_2$ .
- advantage: improved feedback factor (lower noise, higher speed)
- disadvantage: OTA is subjected to input common mode variations



W. Yang et al., "A 3-V 340-mW 14-b 75-MSample/s CMOS ADC With 85-dB SFDR at Nyquist Input", IEEE J. Solid-State Circuits, pp. 1931-1936, Dec. 2001

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## Overview

- Introduction
- Nonidealities
- Bottom Plate Sampling
- **Design Considerations**

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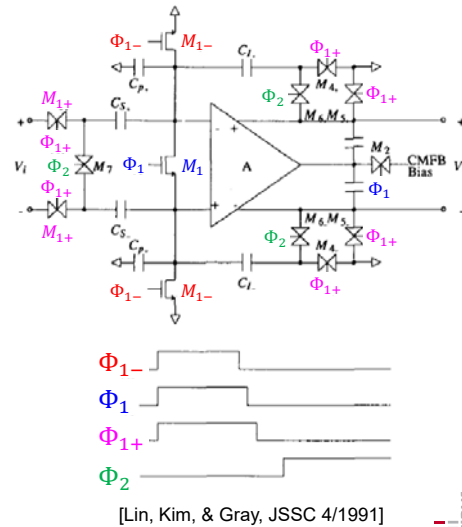
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## Design Considerations I

- $M_{1-}$ -switches only needed to set the common mode in the  $\Phi_1$ -phase,  $M_1$  is the sampling switch  
→ make  $M_1$  larger than  $M_{1-}$
- Ideally, turn off  $M_{1-}$  before  $M_1$ .  
In practice, usually OK to turn off simultaneously.
- total path resistance in track mode:  $R(M_{1+}) + R(M_1)$   
However,  $R(M_{1+})$  is signal dependent.  
→ Make  $R(M_{1+})$  small compared with  $R(M_1)$ .



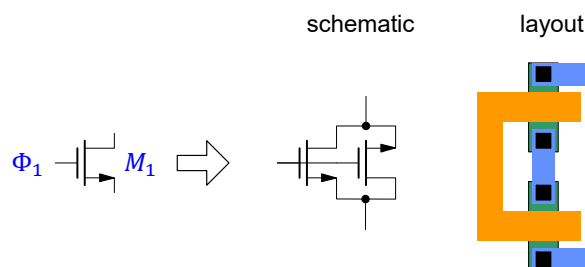
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## Design Considerations II



Use antiparallel devices to implement  $M_1$ .

Guarantees symmetry

- BSIM model not necessarily perfectly symmetric w.r.t. source/drain
  - drain/source asymmetry during fabrication (tilding of wafer)

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## Linearity – Limitations

Linearity of bottom plate sampling circuits is mainly determined by two effects:

- track mode nonlinearity due to  $R_{on} = f(v_{IN})$ 
  - mitigate by clock bootstrapping and proper partitioning of total path resistance
  - most detrimental at high frequencies
- mismatch in half-circuit charge injection due to  $R_{on} = f(v_{IN})$ 
  - Even with clock bootstrapping, the impedance seen by the bottom plate switch may still be input dependent due to back-gate effects → charge injection mismatch
  - often fairly independent of frequency

SFDR of high speed designs

- up to 100 dB at low frequencies
- up to 80 dB up to a few hundreds of Megahertz

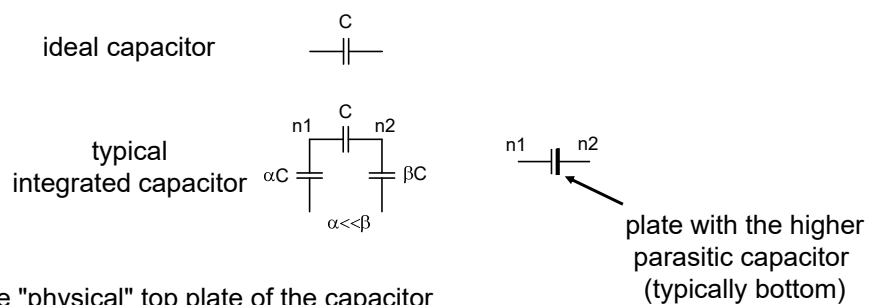
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## Integrated Capacitors



Node n1 is usually the "physical" top plate of the capacitor

- nomenclature is confusing, since this plate is typically used as the "electrical" bottom plate in a bottom plate sampling circuit
- reason: minimize parasitic capacitor at sensitive sampling node in order to minimize substrate coupling

typical values:  $\alpha = 1\%$ ,  $\beta = 10\%$

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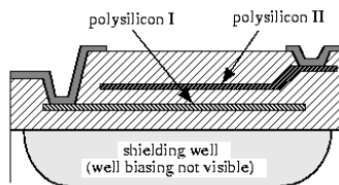
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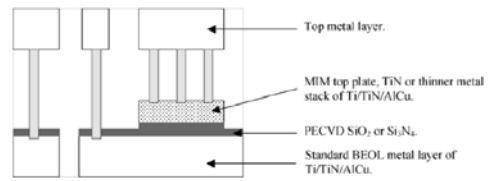


## Typical Capacitors

### Poly-Poly Capacitor



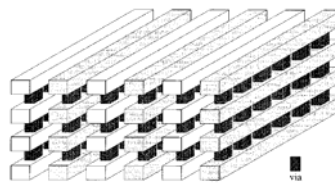
### Metal-Insulator-Metal (MIM)



[Ng, Trans. Electron Dev., 7/2005]

### Vertical Parallel Plate (VPP)

[Aparicio, JSSC 3/2002]



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